# Handwritten Digit Recognition

## 1. Introduction to KNN (K-Nearest Neighbors) for Handwritten Digit Recognition

1) KNN (K-Nearest Neighbor) is a supervised learning method. Its working mechanism is very simple, requiring no training set, and it is one of the simpler classic machine learning algorithms. It can handle regression and classification problems.

2) Methodology

In the feature space, if the majority of the k nearest neighbors (i.e., the closest neighbors in the feature space) of a sample belong to a certain class, then the sample also belongs to that class.

In official terms, the K-Nearest Neighbors algorithm is, given a training dataset, for a new input instance, finding the K nearest neighbors (the K neighbors mentioned above) in the training dataset. If the majority of these K neighbors belong to a certain class, the input instance is classified into that class.

3) Working Principle

There exists a set of sample data, also called the training sample set, and each data point in the sample set has a label, meaning we know the relationship between each data point in the sample set and its corresponding class. After inputting unlabeled data, each feature in the new data is compared with the corresponding features in the sample set, extracting the classification label of the most similar data (nearest neighbor) in the sample set. Generally, we only select the top K most similar data in the sample dataset; this is the origin of K in the K-nearest neighbor algorithm, and K is usually an integer no greater than 20. Finally, the classification of the new data is selected based on the most frequent occurrence among the K most similar data.

4) Advantages and Disadvantages of KNN

- Advantages

Flexible usage, convenient for small sample prediction, high accuracy, insensitive to outliers, no data input assumptions

- Disadvantages

Lacks a training phase, cannot handle multiple samples, high computational and space complexity

5) KNN Implementation Steps:

- Calculate distance

Euclidean distance, i.e.

$$L_2(x_i, x_j) = \left(\sum_{l=1}^{n} |x_i^{(l)} - x_j^{(l)}|^2\right)^{\frac{1}{2}}$$

- Sort points according to increasing distance
- Select the K points with the smallest distance (generally no more than 20)

- Determine the frequency of occurrence of the class of the top K points, frequency = class / k
- Return the class with the highest frequency among the top K points as the predicted classification for the test data

## 2. Introducing the KNN algorithm implementation using handwritten digit recognition as an example

1) Dataset

- Training Set

The training set is pre-classified data. You can refer to ~/content_recognition/KNN/knn-digits/trainingDigits in the directory.

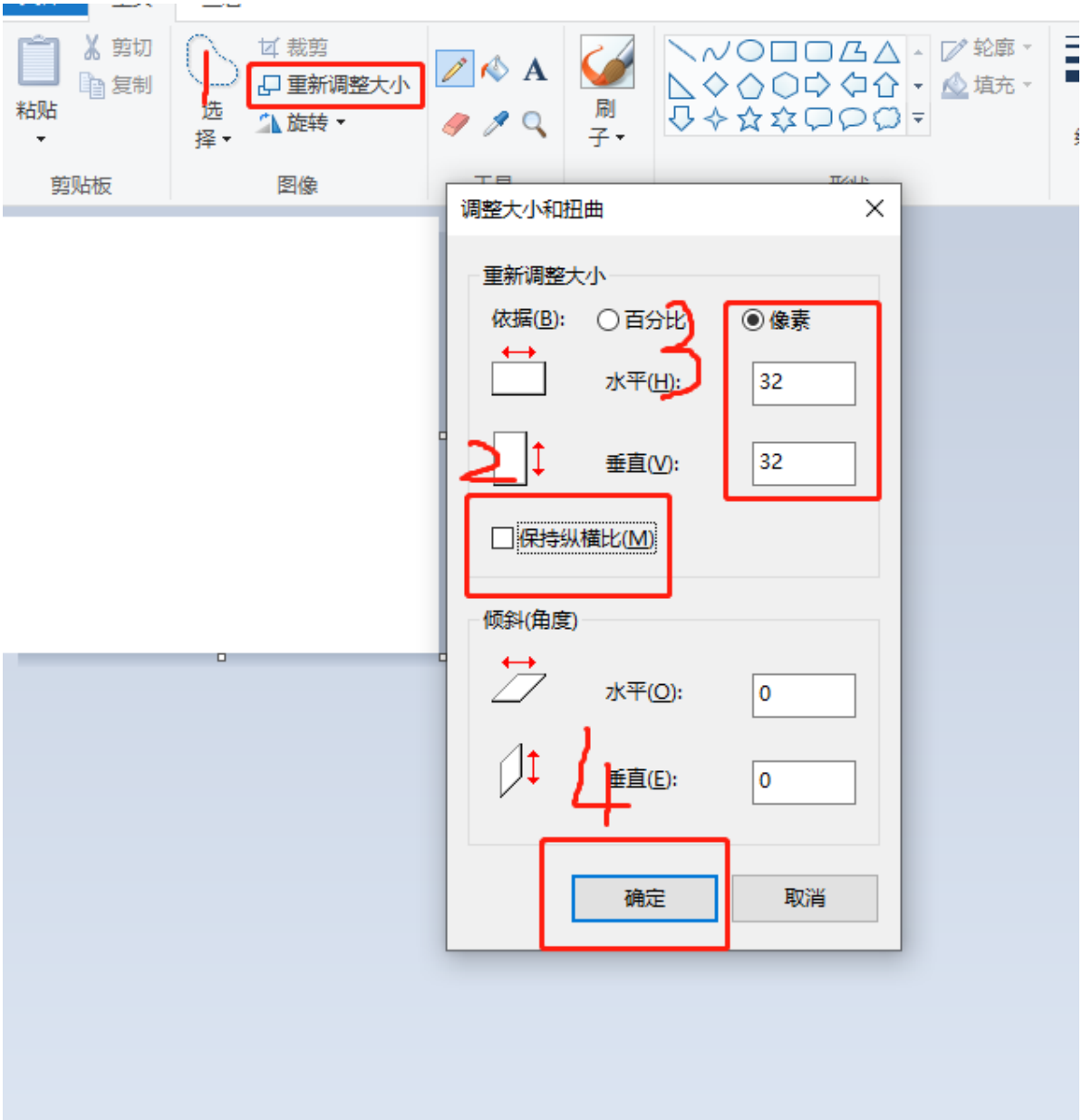
Note: The training set is already trained. If you want to train it yourself, you need to back up the original training set first, as it is quite large.

- Test Set

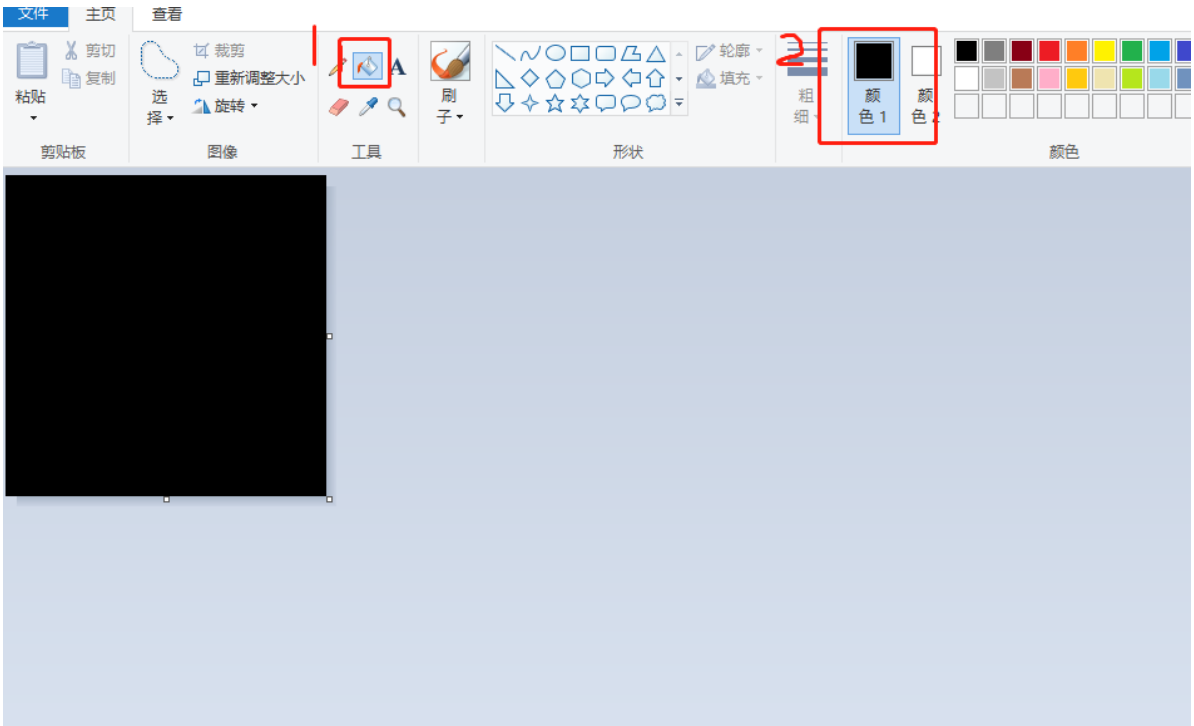
The test set is used to test the algorithm. You can refer to ~/content_recognition/KNN/knn-digits/testDigits in the directory.

2) Creating handwritten digit images using Windows Paint

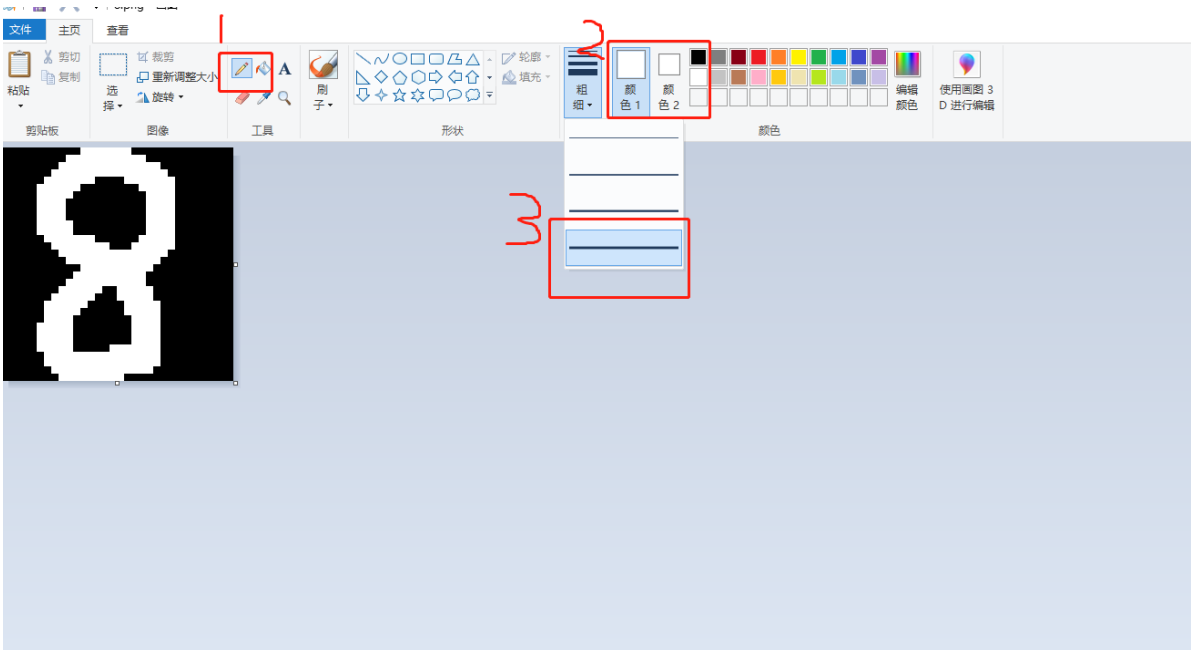
- Open Paint and adjust the resolution to 32*32. Other resolutions are also possible, but it's best to fill the entire area with the drawing; otherwise, the error rate will be high.




- Hold down Ctrl and scroll the mouse wheel upwards to zoom in on the image to its maximum. Select the Paint Bucket tool and choose black to fill the entire image.



- Select the Pencil tool, choose white as the color, and select the maximum line width. As shown in the image below:




After drawing, save it as a PNG image (using 8.png as an example), and copy it to the project directory using WinSCP.

3) Convert the image (.img) to text (.txt)

- Code

Code location: ~/content_recognition/KNN/img2file.py

```
from PIL import Image

import numpy as np


def img2txt(img_path, txt_name):

    im = Image.open(img_path).convert('1').resize((32, 32))  # type:Image.Image

    data = np.asarray(im)

    np.savetxt(txt_name, data, fmt='%d', delimiter='')
img2txt("8.png","./knn-digits/testDigits/8_1.txt")
```

`img_path`: Image filename

`txt_name`: The converted file will be saved in the `~/content_recognition/KNN/knn-digits/testDigits` directory, named `8_1.txt`

- Run the program

```
cd ~/content_recognition/KNN
python img2file.py
```

After the program runs, a file named `8_1.txt` will be generated in the KNN directory. Double-clicking it will show something like this:

```
000000000001111111000000000000000
000000000111111111110000000000000
000000001111111111111100000000000
000000011111111111111110000000000
000000111111100001111111100000000
000001111110000000001111100000000
000001111110000000000111100000000
000001111100000000000111100000000
000001111100000000000111100000000
000001111100000000000111100000000
000001111100000000000111100000000
000001111110000000000111100000000
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000000001111110001111100000000000
000000011111100000111110000000000
000000011111100000011110000000000
000000011111000000011110000000000
000000011110000000011110000000000
000000011110000000011110000000000
000000011110000000011110000000000
000000011110000000011110000000000
000000011110000000011110000000000
000000011110000001111110000000000
000000011111111111111110000000000
000000011111111111111110000000000
000000011111111111111100000000000
000000001111111111111000000000000
```

You can see that the part containing the number 8 roughly encloses the number 8.

4) Run the KNN recognition algorithm program

Open the terminal and run the following command:

```
cd ~/content_recognition/KNN
python knn.py
```

- Code

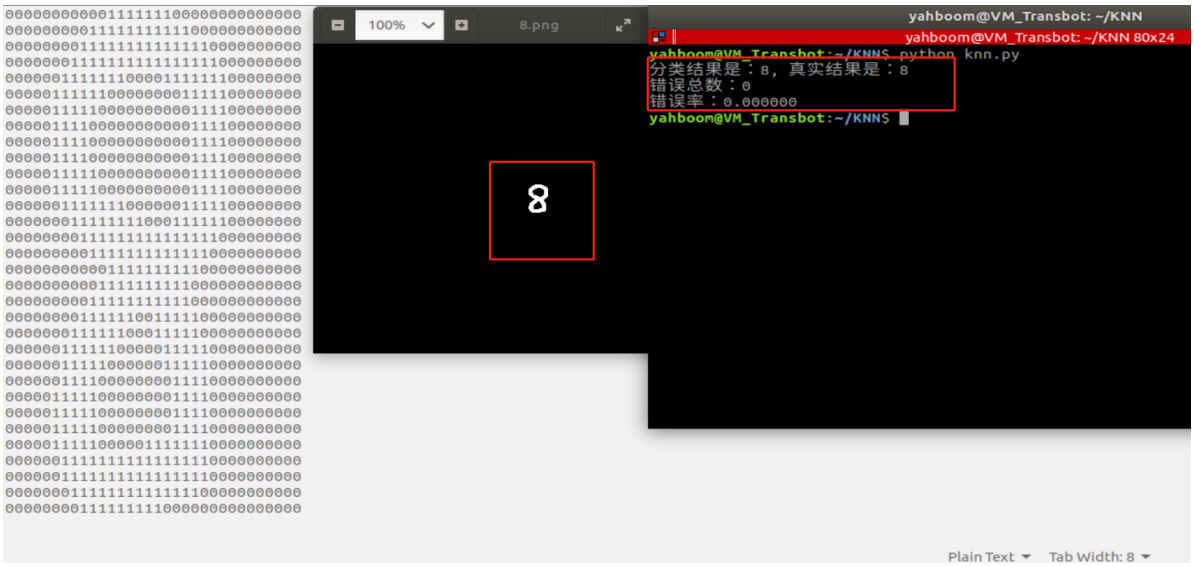
Code location: ~/content_recognition/KNN/knn.py

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Sun Apr 16 14:53:46 2017
```

```
@author: jiangkang
"""

import numpy
import operator
import os
import matplotlib.pyplot as plt

def img2vector(filename):
    returnVect = numpy.zeros((1, 1024))
    file = open(filename)
    for i in range(32):
        lineStr = file.readline()
        for j in range(32):
            returnVect[0, 32 * i + j] = int(lineStr[j])
    return returnVect

def classifier(inX, dataSet, labels, k):
    dataSetSize = dataSet.shape[0]
    diffMat = numpy.tile(inX, (dataSetSize, 1)) - dataSet
    sqDiffMat = diffMat ** 2
    sqDistances = sqDiffMat.sum(axis=1)
    distances = sqDistances ** 0.5
    sortedDistIndicies = distances.argsort()
    classCount = {}
    for i in range(k):
        voteIlabel = labels[sortedDistIndicies[i]]
        classCount[voteIlabel] = classCount.get(voteIlabel, 0) + 1
    sortedClassCount = sorted(classCount.items(), key=operator.itemgetter(1),
reverse=True)
    return sortedClassCount[0][0]
#Train first, then test for recognition. k represents the k-value in the
algorithm - selecting the K most similar data points from the sample dataset. The
k-value is generally an integer not greater than 20.
def handWritingClassTest(k):
    hwLabels = []
    trainingFileList = os.listdir('knn-digits/trainingDigits') #Read training
set data
    m = len(trainingFileList)
    trainingMat = numpy.zeros((m, 1024))
    for i in range(m):
        fileNameStr = trainingFileList[i]
        fileStr = fileNameStr.split('.')[0]
        classNumStr = int(fileStr.split('_')[0])
        hwLabels.append(classNumStr)
        trainingMat[i, :] = img2vector("knn-digits/trainingDigits/%s" %
fileNameStr)
    testFileList = os.listdir('knn-digits/testDigits') #Read test set data
    errorCount = 0.0
    mTest = len(testFileList)
    for i in range(mTest):
        fileNameStr = testFileList[i]
        fileStr = fileNameStr.split('.')[0]
        classNumStr = int(fileStr.split('_')[0])
        vectorTest = img2vector("knn-digits/testDigits/%s" % fileNameStr)
        result = classifier(vectorTest, trainingMat, hwLabels, k)
        print("分类结果是: %d, 真实结果是: %d" % (result, classNumStr))
        if result != classNumStr:
```

```
            errorCount += 1.0
    '''fileStr = "2.txt"
    classNumStr = int(fileStr.split('.')[0])
    vectorTest = img2vector("./2.txt")
    result = classifier(vectorTest, trainingMat, hwLabels, 3)'''
    print("错误总数：%d" % errorCount)
    print("错误率：%f" % (errorCount / mTest))
    return errorCount

handWritingClassTest(1)
```

- Screenshot of the running process




As shown in the figure, the recognized number is 8, which is correct. If the recognition result is different from the actual result, please copy the converted txt document to knn-digits/trainingDigits/ and name it as follows: 8_801.txt, and then retrain and recognize it normally.

- Program description

The text file converted from the image is named as follows, taking 8_1 as an example. knn.py will parse this file name in the program. The underscore is used as the boundary. The 8 at the beginning represents the actual number, and the part after can be customized. See the code.

```
classNumStr = int(fileStr.split('_')[0])
```

Train first, then recognize.